

Applied Machine Learning

Unit 03 — Lecture 03 Notes

Gradient Descent: Stochastic, Mini-Batch and Momentum

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Read this first

These notes are written so that you can learn the material *without* having been in the lecture. Every concept is developed from scratch, every number you see was produced by running the code shown beside it, and every figure was generated from real computation rather than drawn by hand.

Unit II answered *what* quantity a model minimises. This lecture answers *how* the minimising is actually done. The whole subject reduces to one line — *step downhill, again and again* — and everything interesting is in the three questions that line raises: how big a step, computed from how much data, and with how much memory of the last step.

Work through them with a Python session open. You will understand gradient descent far better after computing five steps of it on paper than after reading any number of pages about it.

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1 How to use these notes

Read in order. Section 8 assumes Section 7, which assumes Section 6, which assumes Sections 3 and 4. Nothing here requires anything from Unit II beyond knowing that a loss is a single number we want to make small.

There are four kinds of box:

- **Key idea** — the one sentence to remember a month from now.
- **Worked example** — a calculation done by hand, with small numbers, before any library is used.
- **Caution** — the specific mistake that section exists to prevent.
- **Try it yourself** — something to run, with the expected output given so you can check.

Code appears in a framed block, and directly beneath it a shaded block showing what that code *actually printed* when these notes were compiled. If you run it and get something different, trust your machine and tell me.

1.1 What you need

```
pip install numpy scikit-learn matplotlib
```

Every number in these notes was produced with:

```
import numpy, sklearn, matplotlib
print("numpy", numpy.__version__, "| sklearn", sklearn.__version__,
      "| matplotlib", matplotlib.__version__)
```

```
numpy 2.2.6 | sklearn 1.7.2 | matplotlib 3.10.9
```

Only one dataset is used, and it ships inside scikit-learn: `load_diabetes` — 442 patients, 10 features, a real progression-of-disease measurement as the target. Nothing here downloads anything, so all of it runs on the campus network.

1.2 Learning outcomes

By the end of this lecture you should be able to:

1. State the gradient descent update rule and apply it by hand to a one-parameter function (CO2).
2. Explain how the learning rate controls convergence, derive the stability threshold for a quadratic, and recognise each of the four regimes from a loss curve (CO2).
3. Distinguish batch, stochastic and mini-batch gradient descent by their cost per update, their gradient noise, and their progress per epoch, and justify why mini-batch is the practical default (CO2).
4. State the momentum update, explain it as an exponentially weighted average of past gradients, and predict its behaviour in a ravine and on a well-conditioned problem (CO2).

2 From *what* to *how*

Unit II ended with a complete specification of a learning problem. We have

- a model family with parameters w — say a linear predictor $\hat{y} = x^\top w$;
- a loss that scores one prediction — say squared error;
- a **cost** $J(w)$, the loss averaged over the training set.

The specification is finished. What remains is a purely computational task:

$$\text{find } w^\star = \arg \min_w J(w).$$

Unit III is about the algorithms that do this, and they are the same algorithms whether J came from squared error, cross-entropy or hinge loss. That is the payoff of having separated the two questions.

2.1 Why not just solve it exactly?

For linear regression with squared error, we can. Setting the gradient to zero gives the **normal equations**, $X^\top X w = X^\top y$, with the closed form

$$w^\star = (X^\top X)^{-1} X^\top y.$$

So why is this lecture necessary? Three reasons, each fatal on its own.

1. **Cost.** Inverting $X^\top X$ costs about $O(np^2 + p^3)$. With $p = 10$ that is nothing; with $p = 10^6$ — ordinary for a text model — p^3 is 10^{18} operations, which no computer finishes.
2. **Generality.** The closed form exists only because squared error is quadratic in w . Set the gradient of the logistic loss to zero and you get transcendental equations with no closed-form solution; for a neural network the question is not even well posed.
3. **Memory.** $(X^\top X)^{-1}$ needs all n rows at once. When the data does not fit in RAM — or arrives as a stream — you need a method that can make progress from a fragment.

Key idea

Gradient descent trades an exact answer computed once for an approximate answer improved many times. That trade is what makes it the only optimiser that scales, and it is why every modern machine learning system uses some descendant of it.

We will nevertheless use linear regression as the running example throughout, precisely *because* we can compute the exact answer and check how close the iterative method got. Every measurement in these notes is quoted against the least-squares optimum.

Sources: Zhang, Lipton, Li and Smola, *Dive into Deep Learning*, Ch. 12 (Optimization Algorithms).; Deisenroth, Faisal and Ong, *Mathematics for Machine Learning*, Ch. 7.; Kingma and Ba, “Adam: A Method for Stochastic Optimization,” ICLR 2015.; Ruder, “An overview of gradient descent optimization algorithms,” arXiv:1609.04747, 2016.

3 Gradient descent

3.1 One parameter first

Take a function of a single number, $f(w)$. Its derivative $f'(w)$ answers the question: *if I increase w slightly, does f go up or down, and how fast?*

- $f'(w) > 0$: increasing w increases f . To reduce f , go *left*.
- $f'(w) < 0$: increasing w decreases f . To reduce f , go *right*.

Both cases say: move in the direction of $-f'(w)$. How far? A distance proportional to the slope, scaled by a number $\eta > 0$ that we choose. That is the whole algorithm:

$$w \leftarrow w - \eta f'(w)$$

The number η is the **learning rate** or **step size**. It is the single most consequential hyperparameter in machine learning, and Section 4 is entirely about it.

Note that the step shrinks automatically as we approach a minimum, because the slope there tends to zero. We do not need to slow down by hand; the geometry does it for us.

Five steps of gradient descent, on paper

Minimise $f(w) = (w - 3)^2$, whose minimum is obviously at $w^* = 3$. Take $f'(w) = 2(w - 3)$, start at $w_0 = 0$, and use $\eta = 0.1$.

The update is

$$w \leftarrow w - 0.1 \cdot 2(w - 3) = w - 0.2(w - 3) = 0.8w + 0.6.$$

k	w_k	$f(w_k)$	$f'(w_k)$
0	0.0000	9.0000	-6.0000
1	0.6000	5.7600	-4.8000
2	1.0800	3.6864	-3.8400
3	1.4640	2.3593	-3.0720
4	1.7712	1.5099	-2.4576
5	2.0170	0.9664	

Check the first row yourself: $f'(0) = 2(0 - 3) = -6$, so $w_1 = 0 - 0.1 \times (-6) = 0.6$. Then $f'(0.6) = 2(0.6 - 3) = -4.8$, so $w_2 = 0.6 + 0.48 = 1.08$.

Watch the third column. Every step is smaller than the last, in exactly the ratio 0.8: -6.0, -4.8, -3.84, -3.072. The algorithm is decelerating on its own as it nears the target.

3.2 The error shrinks geometrically

That ratio of 0.8 is not an accident, and it is worth deriving because it explains everything in the next section. Write $e_k = w_k - 3$ for the remaining error. Then

$$e_{k+1} = w_{k+1} - 3 = (w_k - 2\eta(w_k - 3)) - 3 = (1 - 2\eta) e_k.$$

So the error is multiplied by the fixed number $(1 - 2\eta)$ at every step:

$$e_k = (1 - 2\eta)^k e_0, \quad \text{hence} \quad w_k = 3 - 3(1 - 2\eta)^k \quad \text{when } w_0 = 0. \quad (1)$$

With $\eta = 0.1$ the multiplier is 0.8, which is where the pattern in the table comes from. Let us confirm that the closed form matches the iteration.

```
def f(w): return (w - 3.0)**2
def df(w): return 2.0*(w - 3.0)

eta, w = 0.1, 0.0
for k in range(5):
    print("k=%d w=%.4f f(w)=%.4f f'(w)=%+.4f" % (k, w, f(w), df(w)))
    w = w - eta*df(w)
print("k=5 w=%.4f f(w)=%.4f" % (w, f(w)))
```

```
k=0 w=0.0000 f(w)=9.0000 f'(w)=-6.0000
k=1 w=0.6000 f(w)=5.7600 f'(w)=-4.8000
k=2 w=1.0800 f(w)=3.6864 f'(w)=-3.8400
k=3 w=1.4640 f(w)=2.3593 f'(w)=-3.0720
k=4 w=1.7712 f(w)=1.5099 f'(w)=-2.4576
k=5 w=2.0170 f(w)=0.9664
```

```
import numpy as np
k = np.arange(6)
print(np.round(3 - 3*(1 - 2*0.1)**k, 6))
```

```
[0. 0.6 1.08 1.464 1.7712 2.01696]
```

Identical. The iteration and the formula are the same object.

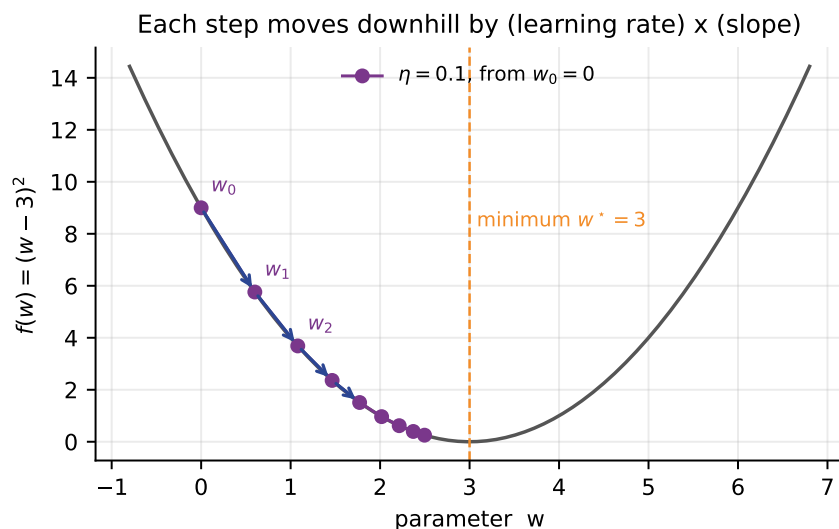


Figure 1: Gradient descent on $f(w) = (w - 3)^2$ from $w_0 = 0$ with $\eta = 0.1$. The arrows mark the first four steps. Points bunch up near the minimum because the slope — and therefore the step — is shrinking.

Key idea

On a quadratic, gradient descent multiplies the remaining error by a fixed factor every iteration. Convergence is **geometric**: each step buys the same *percentage* improvement, not the same absolute improvement. That is why loss curves look like a cliff followed by a long flat tail, and why the last decimal place always costs the most.

3.3 Many parameters

With p parameters, w is a vector and the derivative becomes the **gradient** — the vector of partial derivatives,

$$\nabla J(w) = \left[\frac{\partial J}{\partial w_1}, \frac{\partial J}{\partial w_2}, \dots, \frac{\partial J}{\partial w_p} \right]^\top.$$

The gradient points in the direction of *steepest increase* of J , so $-\nabla J$ points in the direction of steepest decrease. The update is unchanged in form:

$$w \leftarrow w - \eta \nabla J(w). \quad (2)$$

For the mean squared error of a linear model, $J(w) = \frac{1}{n} \sum_i (x_i^\top w - y_i)^2 = \frac{1}{n} \|Xw - y\|^2$, the gradient is

$$\nabla J(w) = \frac{2}{n} X^\top (Xw - y). \quad (3)$$

That single line is the only calculus this lecture needs, and every code example below is a direct transcription of it.

“Steepest descent” is a local claim, not a global one

$-\nabla J$ is the best direction *for an infinitesimally small step*. It is not the direction of the minimum. On a stretched bowl the two can be almost perpendicular — which is exactly the ravine problem that motivates momentum in Section 8.

4 The learning rate

Equation (1) tells us everything about how η behaves on a quadratic. The error multiplier is $|1 - 2\eta|$ for $f(w) = (w - 3)^2$, and in general $|1 - \eta f''|$ where f'' is the curvature. Four regimes follow immediately.

Regime	Multiplier	Behaviour
η too small	just under 1	converges, but takes forever
η well chosen	near 0	converges fast, monotonically
η too large	between -1 and 0	converges, but overshoots and oscillates about the minimum
η past the limit	magnitude > 1	diverges : each step is further out than the last

Setting $|1 - \eta f''| < 1$ gives the **stability threshold**

$$0 < \eta < \frac{2}{L}, \quad L = \text{the largest curvature of } J. \quad (4)$$

For $f(w) = (w - 3)^2$ we have $f'' = 2$, so the threshold is exactly $\eta < 1$. And the multiplier is exactly zero when $\eta = 1/f'' = 0.5$ — gradient descent then lands on the minimum in a *single step*, because for a quadratic that step is Newton’s method.

The four regimes, worked out

Still $f(w) = (w - 3)^2$, still $w_0 = 0$, twenty iterations.

η	$ 1 - 2\eta $	prediction
0.10	0.80	converges slowly, from below
0.50	0.00	exact after one step
0.90	0.80	converges at the same rate as 0.10, but oscillating
1.00	1.00	never converges, never diverges: bounces $0 \rightarrow 6 \rightarrow 0$
1.05	1.10	diverges

The pair $\eta = 0.1$ and $\eta = 0.9$ is the one to think about. Both give $|1 - 2\eta| = 0.8$, so both converge at *exactly the same speed*, one creeping up from below and one flipping from side to side. A learning rate nine times larger bought nothing at all.

```
for eta in [0.1, 0.5, 0.9, 1.0, 1.05]:
    w = 0.0
    for _ in range(20):
        w = w - eta*2.0*(w - 3.0)
    print("eta=%4.2f |1-2*eta| = %.2f w_20 = %12.4f" % (eta, abs(1-2*eta), w))
```

```
eta=0.10 |1-2*eta| = 0.80 w_20 = 2.9654
eta=0.50 |1-2*eta| = 0.00 w_20 = 3.0000
eta=0.90 |1-2*eta| = 0.80 w_20 = 2.9654
eta=1.00 |1-2*eta| = 1.00 w_20 = 0.0000
eta=1.05 |1-2*eta| = 1.10 w_20 = -17.1825
```

Every prediction is confirmed, including the two that look strange. At $\eta = 0.9$ the answer after twenty steps is 2.9654 — the same to four decimals as $\eta = 0.1$. At $\eta = 1.0$ the iterate is back exactly where it started: w alternates 0, 6, 0, 6, ... forever, a perpetual-motion machine that never makes progress. And a five per cent increase past the threshold, from 1.00 to 1.05, turns that into -17.18 and falling.

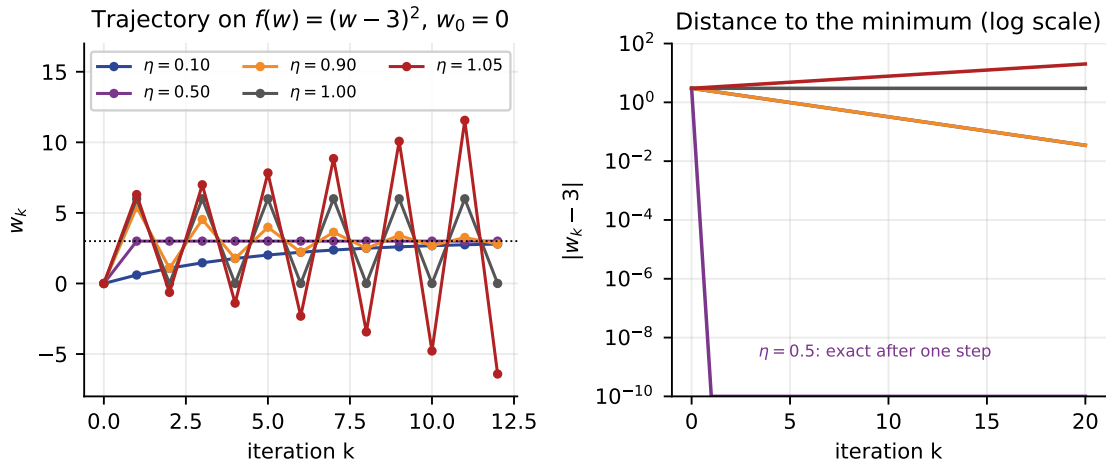


Figure 2: The four regimes. Left: the iterates themselves. Right: distance to the minimum on a logarithmic scale, where geometric convergence is a straight line — downwards for $\eta < 1$, flat for $\eta = 1$, upwards for $\eta = 1.05$. The $\eta = 0.5$ curve falls off the bottom of the axis after one step because the error is exactly zero.

The learning-rate cliff

Convergence does not degrade gracefully as η grows. Up to $2/L$ everything works; past $2/L$ nothing does. There is no warning, and the transition is sharp: in our example $\eta = 1.00$ is neutral and $\eta = 1.05$ is already exploding. This is why a diverging loss is almost always a

learning-rate problem before it is anything else.

4.1 Reading a loss curve

You will not usually know L , so you will diagnose η from the loss curve instead. The four regimes have four distinct signatures:

- **Still falling steeply when you stop** — η too small, or you stopped too early. Multiply η by 3, or train longer.
- **Falls fast, then flattens** — healthy.
- **Falls, then bounces without settling** — η slightly too large, or the gradient noise floor of Section 6.
- **Rises, or becomes nan** — past the threshold. Divide η by 10 and start again.

A good η is a property of the data, not of the algorithm

L depends on how your features are scaled, so a learning rate that works on standardised data can be catastrophic on raw data. This is the single strongest practical argument for the scaling step you will meet in Unit IV, and it is easy to measure — see Section 9.1.

5 Batch gradient descent on real data

Now the same algorithm on the bundled diabetes dataset. We standardise the ten features and the target, which puts the loss at exactly 1.0 when $w = 0$ and makes every number below easy to interpret.

```
from sklearn.datasets import load_diabetes

d = load_diabetes()
X = d.data
X = (X - X.mean(axis=0)) / X.std(axis=0) # standardise the 10 features
y = d.target.astype(float)
y = (y - y.mean()) / y.std() # and the target
n, p = X.shape

H = 2.0/n * (X.T @ X) # Hessian of the mean squared error
ev = np.linalg.eigvalsh(H)
print("examples n, features p ", n, p)
print("largest curvature L %.4f" % ev.max())
print("smallest curvature %.4f" % ev.min())
print("condition number %.1f" % (ev.max()/ev.min()))
print("stability limit 2/L %.4f" % (2/ev.max()))
```

```
examples n, features p 442 10
largest curvature L 8.0484
smallest curvature 0.0171
condition number 470.1
stability limit 2/L 0.2485
```

Three things are now known before we run anything. Any learning rate at or above 0.2485 will diverge. The curvature differs by a factor of 470 between the steepest and the shallowest

direction — remember that number, it is the reason Section 8 exists. And because the loss is quadratic in w , all of this is exact, not a rule of thumb.

```
def mse(X, y, w):
    r = X @ w - y
    return float(np.mean(r**2))

def grad(X, y, w):
    r = X @ w - y
    return 2.0/len(y) * (X.T @ r)

w = np.zeros(p)
print("epoch 0 MSE %.4f" % mse(X, y, w))
for epoch in range(1, 6):
    w = w - 0.005*grad(X, y, w)
    print("epoch %2d MSE %.4f" % (epoch, mse(X, y, w)))
w_star = np.linalg.lstsq(X, y, rcond=None)[0]
print("least-squares optimum MSE %.4f" % mse(X, y, w_star))
```

```
epoch 0 MSE 1.0000
epoch 1 MSE 0.9713
epoch 2 MSE 0.9447
epoch 3 MSE 0.9199
epoch 4 MSE 0.8969
epoch 5 MSE 0.8754
least-squares optimum MSE 0.4823
```

This is what **batch gradient descent** looks like: one update per pass over the data, using the gradient computed from all 442 examples. Five passes have closed about a quarter of the gap between 1.0000 and 0.4823. It works, and it is slow.

Vocabulary, fixed now so it stays fixed

An **epoch** is one complete pass over the training set. An **update** (or iteration, or step) is one application of Equation (2). Batch gradient descent performs exactly one update per epoch. The methods in the next two sections perform many, and that difference is the entire content of Sections 6 and 7.

5.1 The cost of one update

Computing ∇J from Equation (3) touches every row of X : the cost is $O(np)$ per update. With $n = 442$ that is instant. With $n = 10^8$ — a modest industrial dataset — a single step of batch gradient descent means reading a hundred million rows off disk, and you would need thousands of such steps.

That is the observation the rest of this lecture is built on.

6 Stochastic gradient descent

6.1 The structure we have been ignoring

The cost is an *average* over the training set:

$$J(w) = \frac{1}{n} \sum_{i=1}^n \ell_i(w), \quad \text{so} \quad \nabla J(w) = \frac{1}{n} \sum_{i=1}^n \nabla \ell_i(w).$$

Batch gradient descent computes all n terms of that sum and then takes one step. But if you want to estimate an average, you do not need every term — you need a sample. Pick one index i uniformly at random. Then

$$\mathbb{E}_i[\nabla \ell_i(w)] = \sum_{i=1}^n \frac{1}{n} \nabla \ell_i(w) = \nabla J(w). \quad (5)$$

The gradient of a single randomly chosen example is an **unbiased estimate** of the full gradient. It is a very noisy estimate — but it is correct on average, and it costs $1/n$ as much.

Stochastic gradient descent (SGD) takes that estimate and steps with it anyway:

$$\text{pick } i \text{ at random; } \quad w \leftarrow w - \eta \nabla \ell_i(w). \quad (6)$$

In one pass over the data, SGD performs n updates where batch gradient descent performs one — for the same total arithmetic.

Key idea

Batch gradient descent takes one very good step per epoch. SGD takes n mediocre steps per epoch. Because progress compounds and the errors partly cancel, n mediocre steps beat one good step by a wide margin, and the margin grows with n .

6.2 By hand, on three points

One epoch of SGD against one batch step

Three examples, one feature, no intercept: predict $\hat{y} = wx$ with

$$x = (1, 2, 3), \quad y = (2, 4, 7).$$

The per-example loss is $\ell_i(w) = (wx_i - y_i)^2$, so $\nabla \ell_i(w) = 2x_i(wx_i - y_i)$. Start at $w = 0$ with $\eta = 0.05$.

One batch step. The full gradient is the mean of the three:

$$\nabla J(0) = \frac{1}{3} [2(1)(0 - 2) + 2(2)(0 - 4) + 2(3)(0 - 7)] = \frac{1}{3} [-4 - 16 - 42] = -20.6667.$$

$$w \leftarrow 0 - 0.05 \times (-20.6667) = \mathbf{1.0333}.$$

One epoch of SGD, visiting the examples in order:

step	gradient at the current w	value	new w
$x = 1$	$2(1)(0 \cdot 1 - 2)$	-4.0000	0.2000
$x = 2$	$2(2)(0.2 \cdot 2 - 4)$	-14.4000	0.9200
$x = 3$	$2(3)(0.92 \cdot 3 - 7)$	-25.4400	2.1920

The exact least-squares answer is $w^* = \frac{\sum x_i y_i}{\sum x_i^2} = \frac{31}{14} = 2.2143$.

After the same amount of arithmetic — three per-example gradients either way — batch gradient descent is at 1.0333 and SGD is at 2.1920, within one per cent of the exact optimum. Why? Because SGD's second gradient was evaluated at $w = 0.2$ rather than at $w = 0$. It used information the batch method threw away by insisting on finishing the sum first.

```
xs = np.array([1., 2., 3.])
ys = np.array([2., 4., 7.])
```

```

eta = 0.05

w = 0.0
g = np.mean(2*xs*(w*xs - ys))
print("one full-batch step: gradient %+.4f -> w = %.4f" % (g, w - eta*g))

w = 0.0
for xi, yi in zip(xs, ys):
    gi = 2*xi*(w*xi - yi)
    w = w - eta*gi
    print(" SGD on x=%.0f: gradient %+.4f -> w = %.4f" % (xi, gi, w))
print("least-squares optimum w* = %.4f" % (np.sum(xs*ys)/np.sum(xs*xs)))

```

```

one full-batch step: gradient -20.6667 -> w = 1.0333
SGD on x=1: gradient -4.0000 -> w = 0.2000
SGD on x=2: gradient -14.4000 -> w = 0.9200
SGD on x=3: gradient -25.4400 -> w = 2.1920
least-squares optimum w* = 2.2143

```

Shuffle every epoch

The example above visited the points in a fixed order, which is fine for a demonstration and wrong in practice. If your data is sorted by label — and data very often is — fixed-order SGD spends the first half of every epoch pulling the parameters towards one class and the second half pulling them back. Draw a fresh permutation at the start of each epoch. Every implementation below does.

6.3 The price: SGD never quite arrives

Unbiased does not mean noiseless. Near the optimum the full gradient is nearly zero, but the individual $\nabla \ell_i$ are not: they point in all directions and cancel only on average. So SGD with a fixed step size does not converge to w^* . It converges to a *cloud* around w^* whose size grows with η — the **noise floor**.

The classical fix is to shrink the step size over time. The Robbins–Monro conditions say a schedule η_t guarantees convergence if

$$\sum_{t=1}^{\infty} \eta_t = \infty \quad \text{and} \quad \sum_{t=1}^{\infty} \eta_t^2 < \infty.$$

The first condition says the steps must not shrink so fast that you stop before arriving; the second says they must shrink fast enough to damp the noise. $\eta_t = \eta_0/(1 + ct)$ satisfies both. Here is the effect, measured.

```

def run_decay(eta0=0.02, epochs=30, seed=1, decay=True):
    rng = np.random.default_rng(seed); w = np.zeros(p); t = 0
    for _ in range(epochs):
        for i in rng.permutation(n):
            t += 1
            eta = eta0/(1 + 0.002*t) if decay else eta0
            w = w - eta*grad(X[i:i+1], y[i:i+1], w)
    return mse(X, y, w)

print("SGD (B=1, 30 epochs), constant step 0.02 MSE %.4f" % run_decay(decay=False))
print("SGD (B=1, 30 epochs), decayed step MSE %.4f" % run_decay(decay=True))
print("least-squares optimum MSE %.4f" % mse(X, y, w_star))

```

```
SGD (B=1, 30 epochs), constant step 0.02 MSE 0.5395
SGD (B=1, 30 epochs), decayed step MSE 0.4834
least-squares optimum MSE 0.4823
```

With a constant step the run stalls at 0.5395 and stays there — it is not converging slowly, it is orbiting. Decaying the step closes almost the entire remaining gap, reaching 0.4834 against an optimum of 0.4823.

Two different things look the same on a loss curve

A loss that bounces without settling can mean η is above the stability threshold, *or* that you have reached the SGD noise floor. They need opposite responses — reduce η in both cases, but in the first the training is broken and in the second it is finished. Distinguish them by evaluating the loss on the *whole* training set rather than on the current mini-batch: a true divergence rises, a noise floor is flat with jitter.

7 Mini-batch gradient descent

Batch gradient descent uses n examples per update; SGD uses 1. Nothing forces those to be the only options. **Mini-batch gradient descent** uses B examples, with $1 \leq B \leq n$:

$$w \leftarrow w - \eta \frac{1}{B} \sum_{i \in \mathcal{B}} \nabla \ell_i(w), \quad \mathcal{B} \text{ a random subset of size } B. \quad (7)$$

Batch and stochastic gradient descent are the two endpoints, $B = n$ and $B = 1$. In practice the answer is always somewhere in between, and this section is about why.

7.1 How the noise depends on B

Averaging B independent unbiased estimates divides the variance by B , so the *standard deviation* of the mini-batch gradient falls like $1/\sqrt{B}$. That is the central trade-off in one line, and we can measure it exactly.

```
rng = np.random.default_rng(0)
w0 = np.zeros(p)
g_full = grad(X, y, w0)
print("|| full gradient || %.4f" % np.linalg.norm(g_full))
for B in [1, 2, 4, 8, 16, 32, 64, 128, 256]:
    sq = []
    for _ in range(3000):
        idx = rng.choice(n, size=B, replace=False)
        sq.append(np.sum((grad(X[idx], y[idx], w0) - g_full)**2))
    print("B=%4d RMS ||g_B - g|| = %.4f" % (B, np.sqrt(np.mean(sq))))
```

```
|| full gradient || 2.4157
B= 1 RMS ||g_B -g|| = 6.3234
B= 2 RMS ||g_B -g|| = 4.4099
B= 4 RMS ||g_B -g|| = 3.1309
B= 8 RMS ||g_B -g|| = 2.1805
B= 16 RMS ||g_B -g|| = 1.5199
B= 32 RMS ||g_B -g|| = 1.0641
B= 64 RMS ||g_B -g|| = 0.7152
B= 128 RMS ||g_B -g|| = 0.4617
B= 256 RMS ||g_B -g|| = 0.2507
```

Read the first two lines together and be slightly alarmed: the full gradient has length 2.4157, and a single-example gradient is typically wrong by 6.3234 — an error more than *twice the size of the thing being estimated*. A single stochastic step frequently points uphill. It still works, because the errors cancel over many steps.

Now read down the column. Each doubling of B multiplies the error by about $1/\sqrt{2} \approx 0.707$: $6.32 \rightarrow 4.41 \rightarrow 3.13 \rightarrow 2.18 \rightarrow 1.52$. Going from $B = 1$ to $B = 32$ costs 32 times more arithmetic per update and buys a factor of only $6.32/1.06 = 5.9$ — almost exactly $\sqrt{32} = 5.66$.

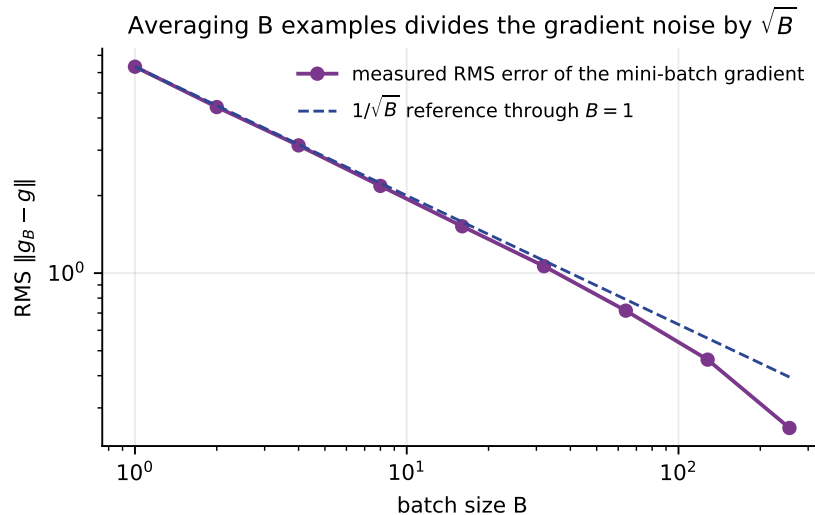


Figure 3: Measured RMS error of the mini-batch gradient against batch size, log–log, with a $1/\sqrt{B}$ reference line. The measured curve falls below the reference at large B because we sample *without* replacement, so a batch of 256 out of 442 already contains most of the data; at $B = n$ the error is exactly zero.

The diminishing return that decides everything

Gradient quality improves as $\sqrt{B}$ while cost grows as B . Doubling the batch size therefore buys you 41% less noise for 100% more work. Large batches are a bad deal for accuracy — which is why nobody uses $B = n$.

7.2 Progress per epoch

If small batches give better value per unit of arithmetic, why not $B = 1$? Run all three for thirty epochs at the same learning rate.

```
def run(B, epochs=30, eta=0.005, seed=1):
    rng = np.random.default_rng(seed)
    w = np.zeros(p)
    hist = [mse(X, y, w)]
    for _ in range(epochs):
        order = rng.permutation(n) if B < n else np.arange(n)
        for s in range(0, n, B):
            idx = order[s:s+B]
            w = w - eta*grad(X[idx], y[idx], w)
            hist.append(mse(X, y, w))
    return np.array(hist)

h_batch, h_mini, h_sgd = run(n), run(32), run(1)
```

```

print(" updates/epoch after 1 epoch after 30 epochs")
print("batch B=442 1 %.4f %.4f" % (h_batch[1], h_batch[30]))
print("mini B= 32 14 %.4f %.4f" % (h_mini[1], h_mini[30]))
print("SGD B= 1 442 %.4f %.4f" % (h_sgd[1], h_sgd[30]))
print("least-squares optimum %.4f" % mse(X, y, w_star))

```

```

          updates/epoch after 1 epoch after 30 epochs
batch B=442 1 0.9713 0.6219
mini B= 32 14 0.7393 0.4860
SGD B= 1 442 0.5268 0.4859
least-squares optimum 0.4823

```

Every one of those rows used the same number of per-example gradient evaluations. After thirty epochs batch gradient descent has reached 0.6219, still a long way from 0.4823, while both stochastic methods have essentially finished. And after a *single* epoch SGD is already at 0.5268 — better than batch gradient descent manages in thirty.

That comparison deserves a number of its own.

```

target = h_sgd[1]
h_long = run(n, epochs=3000)
print("SGD reaches MSE %.4f after a single epoch." % target)
print("Full-batch GD needs %d epochs to reach the same MSE." % np.argmax(h_long <=
    target))

```

```

SGD reaches MSE 0.5268 after a single epoch.
Full-batch GD needs 80 epochs to reach the same MSE.

```

Eighty passes over the data against one. Both spend the same arithmetic per pass. The whole of that factor comes from SGD updating its parameters 442 times per pass instead of once.

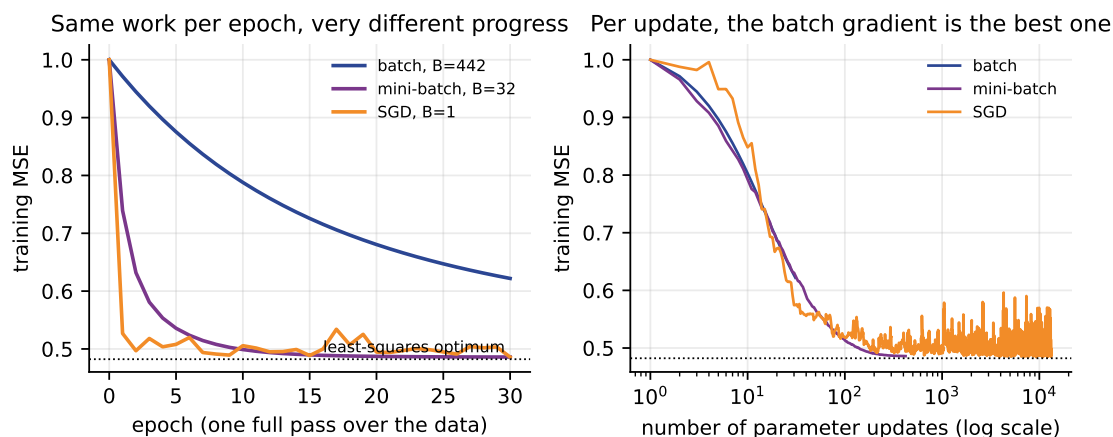


Figure 4: Left: training MSE per epoch — equal work along the horizontal axis, and the small batches win overwhelmingly. Right: the same runs plotted against the number of parameter updates, where the ordering reverses. Per update the batch gradient is the best one; per unit of work it is by far the worst.

The two panels of Figure 4 contradict each other, and both are true. Per *update*, more data is better — the blue curve on the right descends fastest. Per unit of *work*, more data is worse — the same curve on the left descends slowest. Optimisation is judged by the second criterion, because work is what you pay for. The right-hand panel also shows the noise floor plainly: past

about a hundred updates the SGD curve stops descending and becomes a band of jitter around 0.49.

7.3 So why not $B = 1$?

Because arithmetic operations are not the same thing as time. A mini-batch gradient is one matrix–vector product; a hundred single-example gradients are a hundred tiny operations, each carrying the full overhead of a Python-level loop, a memory allocation and a kernel launch.

```
import time
for B in (1, 32, 442):
    best = float("inf")
    for _ in range(25):
        w = np.zeros(p)
        t0 = time.perf_counter()
        for s in range(0, n, B):
            w = w - 0.005*grad(X[s:s+B], y[s:s+B], w)
        best = min(best, time.perf_counter() - t0)
    print("B=%4d %3d updates/epoch %7.3f ms per epoch" % (B, -(n//B), best*1000))
```

```
B= 1 442 updates/epoch 1.224 ms per epoch
B= 32 14 updates/epoch 0.040 ms per epoch
B= 442 1 updates/epoch 0.005 ms per epoch
```

Identical arithmetic; 31 **times** the wall-clock cost for $B = 1$ against $B = 32$ on this machine. And $B = 32$ reached MSE 0.4860 after thirty epochs against SGD’s 0.4859 — statistically the same answer, thirty-one times faster.

These timings are machine-dependent

The ratio, not the absolute times, is the point. On a GPU the effect is far larger: a device with thousands of cores is almost entirely idle processing one example at a time, and batch sizes of 256 or 1024 can cost no more wall-clock time per update than a batch of 32.

7.4 The summary table

	B	updates/epoch	gradient noise	cost/update	vectorises
Batch	n	1	none	$O(np)$	yes
Mini-batch	32–512	n/B	moderate	$O(Bp)$	yes
Stochastic	1	n	large	$O(p)$	no

Why mini-batch wins

Mini-batch gradient descent is not a compromise that is slightly worse than both extremes. It is strictly better than both. It keeps almost all of SGD’s update frequency — 14 updates per epoch instead of 1 was enough to reach the optimum here — while recovering the vectorised arithmetic that makes hardware fast, and cutting the gradient noise by $\sqrt{B}$ into the bargain. It is the default in every deep-learning framework for these reasons and no others.

Terminology in the wild

Almost everything called “SGD” in a paper, a library or a job advert is mini-batch gradient descent. `torch.optim.SGD` takes whatever batch size your data loader produces. Genuine one-example-at-a-time SGD is rare outside online-learning settings. When someone says

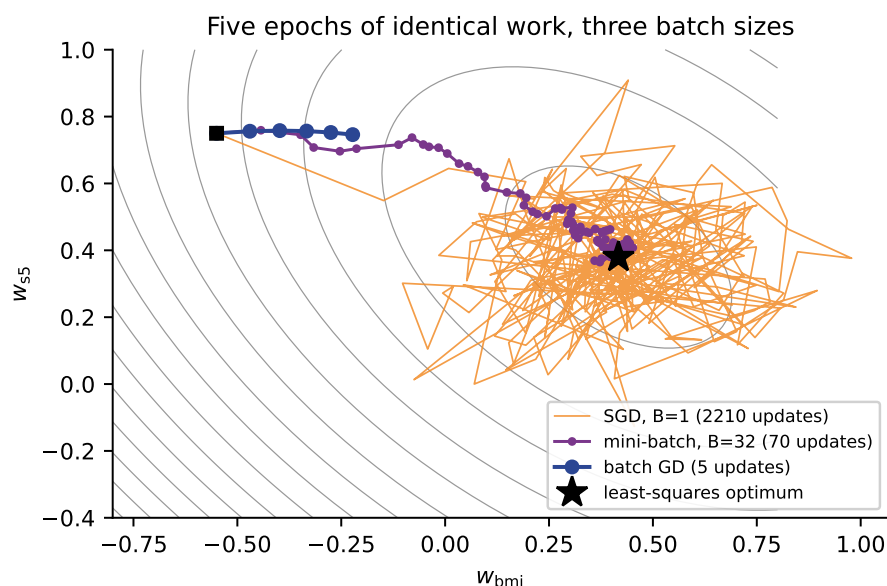


Figure 5: Five epochs of identical work on a two-parameter version of the problem (features `bmi` and `s5`). Batch gradient descent manages five updates and barely leaves the start. Mini-batch takes a fairly direct path in 70 updates. SGD’s 2210 updates arrive quickly and then rattle around the optimum without settling — the noise floor, seen from above.

SGD, assume $B > 1$ unless they say otherwise.

7.5 Choosing B in practice

- Start at $B = 32$: the commonest default, and rarely a bad one.
- Use powers of two — memory alignment and GPU kernels are built around them, and $B = 32$ and $B = 33$ can differ measurably in speed.
- The binding constraint is usually memory, not statistics.
- If you increase B you may increase η , because the gradient is less noisy; a common rule of thumb is to scale η linearly with B .
- Tune η before B . The learning rate matters far more.

Try it yourself

Re-run the `run()` function above with $B = 4, 8, 64, 128$ and plot MSE after five epochs against B . You should find a broad flat region rather than a sharp optimum — which is exactly why $B = 32$ is a safe default and why it is not worth much tuning effort.

8 Momentum

8.1 The problem momentum solves

Recall the number we measured in Section 5: the curvature of our loss differs by a factor of 470 between the steepest and the shallowest direction. Such a loss surface is a **ravine** — a long, narrow valley with steep walls and a gently sloping floor.

Gradient descent handles a ravine badly, and Equation (4) says exactly why. The learning rate must satisfy $\eta < 2/L$ where L is the *largest* curvature, so η is dictated by the steep walls. But the progress that matters is along the floor, where the curvature is $L/470$. The result: the iterates bounce from wall to wall while creeping along the valley. The step size that keeps the crossing direction stable is 470 times too small for the direction we actually care about.

Key idea

In a ravine, gradient descent is limited by the direction you do not want to travel in. Momentum exists to break that coupling.

8.2 The heavy ball

Think of a ball rolling down the valley rather than a hiker taking independent steps. The ball accumulates velocity. Along the floor, where the slope points the same way every step, velocity builds up and the ball accelerates. Across the valley, where the slope reverses every step, the contributions cancel and the oscillation is damped.

Written out, with a **velocity** vector v and a momentum coefficient $\beta \in [0, 1)$:

$$v \leftarrow \beta v + \nabla J(w), \quad w \leftarrow w - \eta v. \quad (8)$$

Setting $\beta = 0$ recovers plain gradient descent exactly. This is Polyak’s heavy-ball method, and it is the form used by `torch.optim.SGD` and by *Dive into Deep Learning* §12.6.

8.3 Momentum is an exponentially weighted average

Unroll the recursion for v , starting from $v_0 = 0$:

$$v_t = \beta v_{t-1} + g_t = g_t + \beta g_{t-1} + \beta^2 g_{t-2} + \cdots = \sum_{j=0}^{t-1} \beta^j g_{t-j}.$$

The velocity is a weighted sum of *all* past gradients, with weights decaying geometrically. Recent gradients dominate; a gradient from 20 steps ago carries weight $0.9^{20} = 0.12$.

The total weight is the geometric series

$$\sum_{j=0}^{\infty} \beta^j = \frac{1}{1 - \beta}.$$

That gives the number worth memorising. If the gradient is roughly constant over many steps — which is precisely the situation along the valley floor — then v settles at about $g/(1 - \beta)$ and the effective step becomes

$$\eta_{\text{eff}} \approx \frac{\eta}{1 - \beta}. \quad (9)$$

With $\beta = 0.9$ that is 10η ; with $\beta = 0.99$ it is 100η . Momentum is a $10\times$ speed-up in consistent directions, obtained without raising η in the inconsistent ones.

Four steps of momentum, on paper

Back to $f(w) = (w - 3)^2$, $\eta = 0.1$, now with $\beta = 0.9$. Start $w_0 = 0$, $v_0 = 0$, and recall $g = 2(w - 3)$.

k	g_k	$v_k = 0.9v_{k-1} + g_k$	$w_k = w_{k-1} - 0.1v_k$
1	-6.0000	$0 - 6.0000 = -6.0000$	$0 + 0.6000 = 0.6000$
2	-4.8000	$-5.4000 - 4.8000 = -10.2000$	$0.6 + 1.0200 = 1.6200$
3	-2.7600	$-9.1800 - 2.7600 = -11.9400$	$1.62 + 1.1940 = 2.8140$
4	-0.3720	$-10.7460 - 0.3720 = -11.1180$	$2.814 + 1.1118 = \mathbf{3.9258}$

Compare with plain gradient descent, which was at 1.7712 after four steps. Momentum is at 3.9258 — it has already gone *past* the minimum at 3.

Look at row 4 to see how. The gradient there is nearly zero (-0.372 , because w is close to 3), yet the velocity is still -11.12 , carrying nine tenths of everything accumulated earlier. The ball arrives at the bottom of the bowl travelling fast and rolls up the far side.

```
eta, beta = 0.1, 0.9
w, v = 0.0, 0.0
for k in range(1, 5):
    g = 2.0*(w - 3.0)
    v = beta*v + g
    w = w - eta*v
    print("k=%d g=%+8.4f v=%+9.4f w=%.4f" % (k, g, v, w))
```

```
k=1 g= -6.0000 v= -6.0000 w=0.6000
k=2 g= -4.8000 v= -10.2000 w=1.6200
k=3 g= -2.7600 v= -11.9400 w=2.8140
k=4 g= -0.3720 v= -11.1180 w=3.9258
```

8.4 Yes, momentum overshoots

Overshooting is not a bug to be tuned away; it is inherent to the method, and whether it is worth paying for depends entirely on the shape of the problem. Here is the full picture on the same one-dimensional quadratic.

```
for beta in (0.0, 0.5, 0.9):
    w, v, hit, top = 0.0, 0.0, None, 0.0
    for k in range(1, 201):
        v = beta*v + 2.0*(w - 3.0)
        w = w - 0.1*v
        top = max(top, w)
        if hit is None and abs(w - 3.0) < 1e-3:
            hit = k
    print("beta=%.1f first k with |w-3|<1e-3: %3d highest w reached: %.4f"
          % (beta, hit, top))
```

```
beta=0.0 first k with |w-3|<1e-3: 36 highest w reached: 3.0000
beta=0.5 first k with |w-3|<1e-3: 20 highest w reached: 3.2092
beta=0.9 first k with |w-3|<1e-3: 92 highest w reached: 5.1269
```

Three different answers, and the middle one is the interesting one.

- $\beta = 0$: 36 steps, never overshoots. Plain gradient descent approaches monotonically from below.

- $\beta = 0.5$: 20 steps, overshoots to 3.2092. Nearly twice as fast, at the cost of a small excursion past the target.
- $\beta = 0.9$: 92 steps, overshoots to 5.1269 — it travels past the minimum at 3 almost as far as it started from it. Two and a half times *slower* than plain gradient descent.

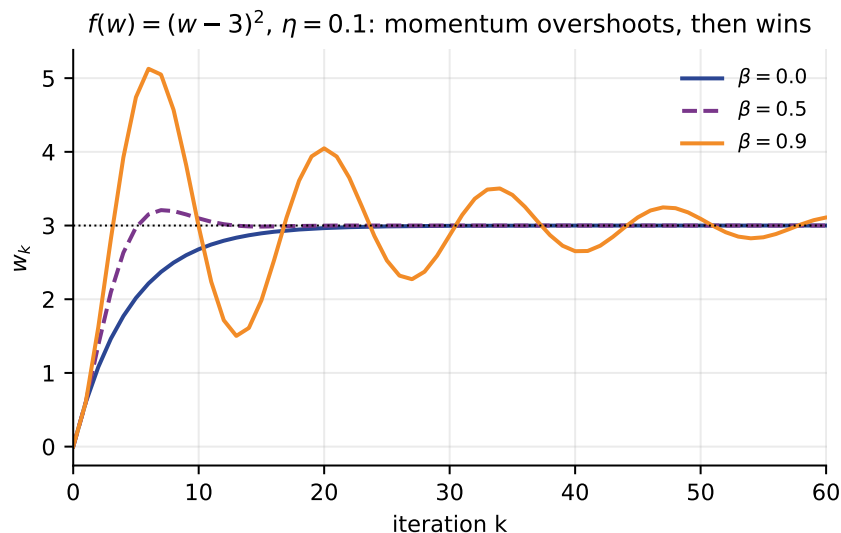


Figure 6: Momentum on $f(w) = (w - 3)^2$ with $\eta = 0.1$. Plain gradient descent (blue) climbs monotonically to 3. $\beta = 0.5$ (purple) gets there faster with a small overshoot. $\beta = 0.9$ (orange) sails up to 5.13 and spends the next eighty iterations ringing down.

Momentum does not help on a well-conditioned problem

A one-dimensional quadratic has condition number 1: there is no ravine, no direction whose oscillation needs damping, and nothing for the accumulated velocity to accelerate along. All momentum can do is overshoot. Setting $\beta = 0.9$ here made the optimiser **two and a half times slower**.

Momentum is not a free improvement you sprinkle on. It is a targeted remedy for badly conditioned problems — which happens to describe almost every real model, but not this one.

8.5 Now put it in a ravine

Here is a quadratic built so we know its condition number exactly:

$$f(w) = 0.01 w_1^2 + 2 w_2^2, \quad \nabla f(w) = (0.02 w_1, 4 w_2), \quad \kappa = \frac{4}{0.02} = 200.$$

The stability ceiling for plain gradient descent is $2/L = 2/4 = 0.5$. We use $\eta = 0.475$, which is 95% of it — as aggressive as plain gradient descent is allowed to be.

```
A = np.array([0.02, 4.0]) # Hessian diagonal: kappa = 200

def run_ravine(eta, beta, iters=1600):
    w = np.array([9.0, 1.0]); v = np.zeros(2)
    for k in range(1, iters+1):
        v = beta*v + A*w
        w = w - eta*v
        if not np.all(np.isfinite(w)) or np.linalg.norm(w) > 1e8:
```

```

        return "diverged at k=%d" % k
    if np.linalg.norm(w) < 1e-3:
        return k
    return "not reached in %d" % iters

print("plain-GD ceiling 2/L = %.3f" % (2/A.max()))
print("momentum ceiling 2(1+beta)/L = %.3f (beta=0.9)" % (2*1.9/A.max()))
for beta in (0.0, 0.5, 0.9):
    print("eta=0.475 beta=%.1f iterations to ||w||<1e-3: %s"
          % (beta, run_ravine(0.475, beta)))
for beta in (0.0, 0.9):
    print("eta=0.900 beta=%.1f iterations to ||w||<1e-3: %s"
          % (beta, run_ravine(0.900, beta)))

```

```

plain-GD ceiling 2/L = 0.500
momentum ceiling 2(1+beta)/L = 0.950 (beta=0.9)
eta=0.475 beta=0.0 iterations to ||w||<1e-3: 954
eta=0.475 beta=0.5 iterations to ||w||<1e-3: 467
eta=0.475 beta=0.9 iterations to ||w||<1e-3: 135
eta=0.900 beta=0.0 iterations to ||w||<1e-3: diverged at k=20
eta=0.900 beta=0.9 iterations to ||w||<1e-3: 161

```

Same learning rate, same starting point, same problem. Plain gradient descent needs 954 iterations; $\beta = 0.9$ needs 135. That is a $7.1\times$ **speed-up** from one extra line of code and one extra vector of state.

The last two lines make a second, independent point. At $\eta = 0.9$ plain gradient descent blows up within twenty iterations, because 0.9 is well past its ceiling of 0.5. Momentum survives, because its stability ceiling is

$$\eta < \frac{2(1+\beta)}{L}, \quad (10)$$

which for $\beta = 0.9$ is 0.95. Momentum does not merely use the permitted learning rates better — it enlarges the set of permitted learning rates by a factor of $1 + \beta$.

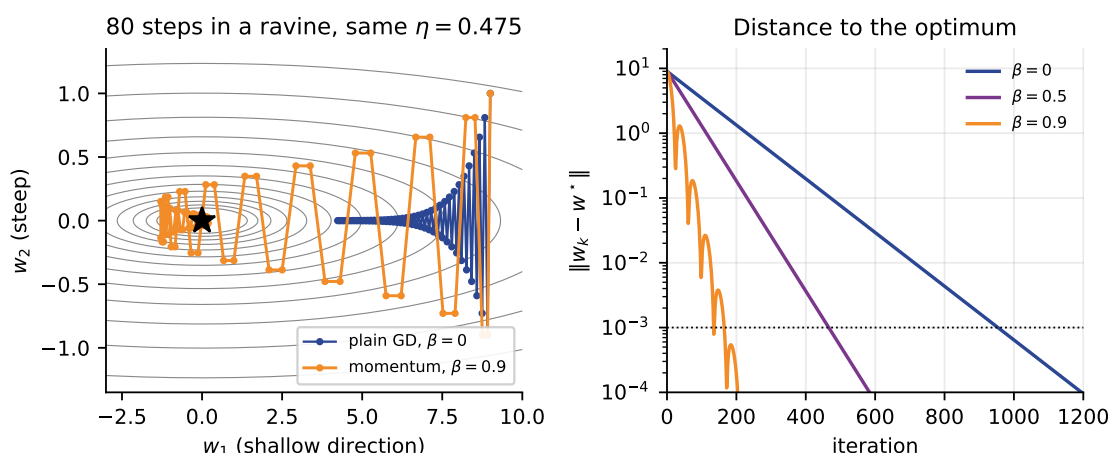


Figure 7: Left: eighty steps in the ravine at $\eta = 0.475$. Plain gradient descent (blue) oscillates across the narrow direction and creeps along the floor, still near $w_1 = 7$ after eighty steps. Momentum (orange) oscillates too — more violently, in fact — but travels the length of the valley and reaches the optimum. Right: distance to the optimum on a log scale, where the slope of each line is the convergence rate.

Figure 7 is worth a slow look, because the left panel does not show what students usually expect. Momentum's path is *messier* than plain gradient descent's, not smoother. What it is not, is

stuck. The lesson is that momentum trades cross-valley tidiness for along-valley speed, and in a ravine that is a trade you always want.

8.6 The same thing on the real dataset

The diabetes least-squares problem had condition number 470 — worse than our synthetic ravine. Momentum should therefore help more, and it does.

```
eta = 0.95*2/ev.max()
for beta in (0.0, 0.9):
    w, v = np.zeros(p), np.zeros(p)
    for k in range(1, 20001):
        v = beta*v + grad(X, y, w)
        w = w - eta*v
        if np.linalg.norm(w - w_star) < 1e-3:
            break
    print("beta=%.1f eta=%.4f full-batch iterations to the optimum: %5d"
          % (beta, eta, k))
```

```
beta=0.0 eta=0.2361 full-batch iterations to the optimum: 1605
beta=0.9 eta=0.2361 full-batch iterations to the optimum: 128
```

1605 iterations down to 128: a $12.5\times$ speed-up on real data, at the same learning rate, from adding momentum. Notice that the speed-up (12.5) is larger than the one on the synthetic ravine (7.1), and that the condition number is larger too (470 against 200). That is the general pattern: the worse the conditioning, the more momentum is worth.

8.7 Momentum with mini-batches

In practice momentum is combined with mini-batch gradient descent, not with full-batch. The update is identical — substitute the mini-batch gradient for ∇J in Equation (8) — and it now does two jobs at once: it accelerates along consistent directions *and* it averages away sampling noise, since the exponentially weighted average of $1/(1 - \beta) = 10$ recent mini-batch gradients is itself a lower-variance estimate.

```
def run_mom(B, beta, eta, epochs=5, seed=1):
    rng = np.random.default_rng(seed)
    w, v = np.zeros(p), np.zeros(p)
    for _ in range(epochs):
        order = rng.permutation(n) if B < n else np.arange(n)
        for s in range(0, n, B):
            idx = order[s:s+B]
            v = beta*v + grad(X[idx], y[idx], w)
            w = w - eta*v
    return mse(X, y, w)

print("mini-batch B=32, eta=0.005, 5 epochs")
for beta in (0.0, 0.9, 0.99):
    print(" beta=%.2f MSE %.4f" % (beta, run_mom(32, beta, 0.005)))
print("least-squares optimum MSE %.4f" % mse(X, y, w_star))
```

```
mini-batch B=32, eta=0.005, 5 epochs
beta=0.00 MSE 0.5358
beta=0.90 MSE 0.4870
beta=0.99 MSE 0.5674
least-squares optimum MSE 0.4823
```

Read the three rows carefully. Going from $\beta = 0$ to $\beta = 0.9$ moves the five-epoch result from 0.5358 to 0.4870 — most of the remaining gap to the optimum, closed by one line. Going on to $\beta = 0.99$ makes things *worse* than no momentum at all, at 0.5674: with an effective learning rate of 100η the run is now overshooting badly.

When you raise β , lower η

Equation (9) says the effective step is $\eta/(1 - \beta)$. Moving β from 0.9 to 0.99 multiplies it by ten. If you change β and keep η , you have changed your learning rate by an order of magnitude without meaning to, and the run above shows what that costs. As a rule of thumb, scale η by $(1 - \beta_{\text{new}})/(1 - \beta_{\text{old}})$.

Two conventions, and they differ by a factor of $1 - \beta$

Some references and libraries write the velocity update as

$$v \leftarrow \beta v + (1 - \beta) g$$

instead of Equation (8). This is a true running average, so v has the same magnitude as g rather than roughly $1/(1 - \beta)$ times it. Andrew Ng’s course and the Adam paper use this form; PyTorch and *Dive into Deep Learning* use ours.

Both are correct, and they produce identical trajectories provided η is scaled by $1/(1 - \beta)$ to match. Move code between them without rescaling and your learning rate silently changes by a factor of ten. Always check which convention a formula is written in before copying a value of η out of it.

8.8 Choosing β

- $\beta = 0.9$ is the default, and it is a good one. It corresponds to averaging roughly the last ten gradients.
- $\beta = 0.99$ averages the last hundred. Use it only on very noisy problems, and reduce η tenfold when you do.
- $\beta = 0$ is not silly on a well-conditioned problem, as Section 8 demonstrated.
- β matters far less than η . Tune the learning rate first, then try 0.9, and stop.

Momentum in one sentence

Momentum replaces the current gradient with an exponentially weighted average of recent gradients, which cancels the components that keep changing sign and amplifies the components that do not — turning a 954-iteration crawl along a ravine into a 135-iteration run.

Momentum treats every coordinate identically: one η , one β , applied to the whole vector. L4 asks what happens when each coordinate gets its own step size, adapted from the gradients that coordinate has seen — Adagrad, RMSprop and Adam.

9 Putting it together

9.1 Feature scaling is an optimisation decision

Section 4 claimed that the usable learning rate is a property of your data. Here is the measurement. We take the diabetes features and multiply one of them by 1000 — an entirely realistic

accident, the difference between recording a quantity in kilometres and in metres.

```
Xbad = load_diabetes().data.copy()
Xbad[:, 0] *= 1000.0 # one feature on a wildly different scale
Xbad = Xbad - Xbad.mean(axis=0)
evb = np.linalg.eigvalsh(2.0/n * (Xbad.T @ Xbad))
print("condition number, standardised %10.1f" % (ev.max()/ev.min()))
print("condition number, one feature x1000 %10.3e" % (evb.max()/evb.min()))
print("stability limit 2/L, standardised %10.4f" % (2/ev.max()))
print("stability limit 2/L, unscaled %10.3e" % (2/evb.max()))
```

```
condition number, standardised 470.1
condition number, one feature x1000 1.168e+08
stability limit 2/L, standardised 0.2485
stability limit 2/L, unscaled 4.420e-04
```

The condition number rises from 470 to 1.2×10^8 , and the largest usable learning rate falls by a factor of 560. Since the number of iterations grows with the condition number, this single unnoticed unit change turns a problem that gradient descent solves in a few hundred iterations into one it cannot solve at all in any reasonable time.

Key idea

Standardising your features is not cosmetic and it is not only about interpretability. It is the cheapest possible improvement to the conditioning of your optimisation problem, and on gradient-based methods it is often the difference between converging and not.

9.2 A recipe

1. **Standardise the features.** Always, before anything else.
2. **Choose $B = 32$** (or the largest power of two that fits comfortably in memory).
3. **Find η by bisection on a log scale.** Try $10^{-1}, 10^{-2}, 10^{-3}$ for a few epochs each. Take the largest that does not diverge, then back off by a factor of three.
4. **Add momentum, $\beta = 0.9$.** If you had already tuned η without momentum, divide it by ten first.
5. **Watch the loss curve** and match it against the four signatures in Section 4.
6. **Decay η towards the end** if the curve flattens into a noisy band above where you expect it to finish.

9.3 What scikit-learn gives you

`SGDRegressor` and `SGDClassifier` implement exactly what is in this lecture, and their arguments are the vocabulary of these notes: `eta0` is η , `learning_rate` selects the decay schedule, `max_iter` counts epochs, `shuffle` draws the permutation each epoch, and `tol` with `n_iter_no_change` implements early stopping on the noise floor. There is no `batch_size`: the implementation is genuinely $B = 1$, which is why the documentation insists you scale your features and warns that results are sensitive to `eta0`. Every one of those design choices is a consequence of something measured above.

10 Summary

1. Gradient descent is $w \leftarrow w - \eta \nabla J(w)$: step in the direction of steepest descent, by an amount proportional to the slope. On a quadratic it multiplies the error by a fixed factor each iteration, so convergence is geometric.
2. The learning rate has a hard ceiling, $\eta < 2/L$, where L is the largest curvature. Below it, everything works; above it, nothing does. In our example $\eta = 1.00$ oscillated forever and $\eta = 1.05$ diverged to -17.18 in twenty steps.
3. Exactly one η is optimal for a quadratic, $\eta = 1/L$, and it solves the problem in a single step. $\eta = 0.1$ and $\eta = 0.9$ converged at identical rates, one monotonically and one oscillating.
4. The cost is an average over examples, so a random example's gradient is an *unbiased* estimate of the full gradient at $1/n$ the cost. That is SGD.
5. SGD is very noisy: on the diabetes data a single-example gradient was typically wrong by 6.32 against a true gradient of length 2.42. It works because the errors cancel.
6. Per epoch, small batches win overwhelmingly. SGD reached MSE 0.5268 in one epoch; full-batch gradient descent needed 80 **epochs** for the same result with the same arithmetic.
7. Mini-batch noise falls as $1/\sqrt{B}$ while cost grows as B : measured, $B=1$ gave 6.32 and $B=32$ gave 1.06, a factor of 5.9 against $\sqrt{32} = 5.66$.
8. Mini-batch wins in practice because it vectorises. $B = 32$ reached the same MSE as $B = 1$ and ran 31 times faster per epoch on the same machine.
9. Momentum, $v \leftarrow \beta v + \nabla J$, $w \leftarrow w - \eta v$, is an exponentially weighted average of past gradients with effective step $\eta/(1 - \beta)$.
10. Momentum overshoots — $\beta = 0.9$ carried a run from 0 past a minimum at 3 all the way to 5.13 — and on a well-conditioned problem it is *slower* than plain gradient descent.
11. In a ravine it is transformative: 954 iterations became 135 ($\kappa = 200$, synthetic) and 1605 became 128 ($\kappa = 470$, real data). It also raises the stability ceiling from $2/L$ to $2(1 + \beta)/L$.
12. Standardising features cut the condition number of the real problem from 1.2×10^8 to 470, and raised the usable learning rate by a factor of 560. Conditioning is something you control.

11 Practice questions

Attempt these before reading Section 12.

1. Minimise $f(w) = (w - 3)^2$ starting from $w_0 = 0$ with $\eta = 0.25$. Compute w_1, w_2, w_3 by hand. What is the error multiplier, and how many steps would you need for $|w_k - 3| < 0.01$?
2. For $f(w) = 5w^2$, find (a) the largest learning rate that converges, (b) the learning rate that reaches the minimum in one step, and (c) the behaviour at $\eta = 0.25$.
3. You have $n = 1,000,000$ training examples. State the number of parameter updates per epoch and the cost of one update for batch gradient descent, for SGD, and for mini-batch with $B = 250$. Which will have made the most progress after reading the dataset once, and why?
4. We measured the RMS mini-batch gradient error as 1.5199 at $B = 16$. Predict the value at $B = 64$ from theory. The measured value is 0.7152. Explain the discrepancy.
5. SGD with a constant step size stalled at MSE 0.5395 while the optimum was 0.4823, and it stayed there for twenty more epochs. Explain what is happening, and give two different fixes.
6. Take $f(w) = (w - 3)^2$, $\eta = 0.1$, $\beta = 0.9$, $w_0 = 0$, $v_0 = 0$. Compute $v_1, w_1, v_2, w_2, v_3, w_3$ by hand.
7. A colleague's model trains well with $\eta = 0.01$ and $\beta = 0.9$. They read that $\beta = 0.99$ is better for noisy problems and change only β . Predict what happens and give the value of η they should have used instead.
8. On a one-dimensional quadratic we measured 36 iterations for $\beta = 0$ and 92 for $\beta = 0.9$. Momentum made it worse. Explain why, and state the property of a problem that determines whether momentum will help.
9. A student ports a training loop from a PyTorch tutorial to code following Andrew Ng's lecture notes, keeping $\eta = 0.001$ and $\beta = 0.9$. Training becomes ten times slower to converge. Diagnose the bug precisely.
10. Multiplying one feature by 1000 raised the condition number from 470 to 1.17×10^8 . Explain, using Equation (4) and the geometric convergence rate, what that does to (a) the largest usable learning rate and (b) the number of iterations required. Why does this make standardisation an optimisation decision rather than a cosmetic one?

12 Solutions

1. The update is $w \leftarrow w - 0.25 \cdot 2(w - 3) = 0.5w + 1.5$. So $w_1 = 1.5$, $w_2 = 2.25$, $w_3 = 2.625$. The error multiplier is $|1 - 2\eta| = 0.5$, so $e_k = 3 \cdot 0.5^k$. We need $3 \cdot 0.5^k < 0.01$, i.e. $0.5^k < 1/300$, i.e. $k > \log(1/300)/\log(0.5) = 8.23$, so $k = 9$ steps. Compare $\eta = 0.1$, whose multiplier of 0.8 needs $k > \log(1/300)/\log(0.8) = 25.6$, so 26 steps — nearly three times as many.

2. $f'(w) = 10w$, $f''(w) = 10$, so $L = 10$. (a) Convergence requires $|1 - 10\eta| < 1$, i.e. $0 < \eta < 0.2$. (b) The multiplier is zero at $\eta = 1/L = 0.1$: one step from anywhere lands exactly on $w = 0$. (c) At $\eta = 0.25$ the multiplier is $|1 - 2.5| = 1.5 > 1$, so the iterates diverge, flipping sign each step: from $w_0 = 1$ you get $-1.5, 2.25, -3.375, \dots$

3.

	updates/epoch	cost per update	
Batch	1	$O(10^6 p)$	one step per pass
SGD	10^6	$O(p)$	one step per example
Mini-batch, $B=250$	4000	$O(250p)$	

All three do the same $O(10^6 p)$ arithmetic per epoch. After one pass, batch gradient descent has taken a single step and will have barely moved from its initialisation. SGD and mini-batch have taken 10^6 and 4000 steps respectively and will be close to convergence. Mini-batch will very likely be *fastest in wall-clock time*, because its 4000 updates are vectorised matrix operations while SGD's million are not — the same effect that made $B=32$ run 31 times faster than $B=1$ in Section 7.

4. The error falls as $1/\sqrt{B}$, so quadrupling B from 16 to 64 should halve it: predicted $1.5199/2 = 0.7600$. The measured value, 0.7152, is about 6% lower.

The reason is that we sample *without* replacement from a finite population of $n = 442$. The variance of the mean of B draws is then $\frac{\sigma^2}{B} \cdot \frac{n-B}{n-1}$, so the standard deviation carries an extra finite-population factor $\sqrt{(n-B)/(n-1)}$. That factor is $\sqrt{426/441} = 0.9829$ at $B = 16$ and $\sqrt{378/441} = 0.9258$ at $B = 64$, so the ratio we should expect is

$$\frac{1}{2} \times \frac{0.9258}{0.9829} = 0.4710, \quad 1.5199 \times 0.4710 = 0.7159,$$

against a measured 0.7152. The effect grows with B and reaches its extreme at $B = n$, where the mini-batch gradient *is* the full gradient and the error is exactly zero.

5. This is the SGD **noise floor**, not slow convergence. Near the optimum the full gradient is nearly zero but individual per-example gradients are not, so each step injects noise of a size set by η . The iterates settle into a stationary cloud around w^* rather than converging to it. Two fixes: (i) **decay the learning rate** on a Robbins–Monro schedule such as $\eta_t = \eta_0/(1 + ct)$ — this took our run from 0.5395 to 0.4834; (ii) **increase the batch size**, which divides the noise by $\sqrt{B}$ — $B = 32$ reached 0.4860 at the same η . Both work; increasing B is usually cheaper because it also vectorises.

6. With $g_k = 2(w_{k-1} - 3)$:

$$g_1 = -6, \quad v_1 = 0.9(0) + (-6) = -6, \quad w_1 = 0 - 0.1(-6) = 0.6;$$

$$g_2 = 2(0.6 - 3) = -4.8, \quad v_2 = 0.9(-6) - 4.8 = -10.2, \quad w_2 = 0.6 + 1.02 = 1.62;$$

$$g_3 = 2(1.62 - 3) = -2.76, \quad v_3 = 0.9(-10.2) - 2.76 = -11.94, \quad w_3 = 1.62 + 1.194 = 2.814.$$

Plain gradient descent reached 1.464 in three steps; momentum reached 2.814. Note that $|v_3|$ exceeds $|g_3|$ by more than four times — the accumulated history now dominates the current gradient entirely.

7. The effective step size is $\eta/(1-\beta)$. At $\beta = 0.9$ that is $0.01/0.1 = 0.1$; at $\beta = 0.99$ it becomes $0.01/0.01 = 1.0$ — ten times larger. They have unintentionally multiplied their learning rate by 10, and the run will overshoot, oscillate, and quite possibly diverge. This is exactly the $\beta = 0.99$ row in Section 8, where the MSE went from 0.4870 to 0.5674, worse than no momentum. To hold the effective step fixed they should have used $\eta = 0.01 \times (1 - 0.99)/(1 - 0.9) = 0.001$.

8. A one-dimensional quadratic has condition number 1: every direction has the same curvature. Momentum’s benefit comes from cancelling oscillation in high-curvature directions while accelerating in low-curvature ones, and with only one direction there is nothing to trade off. All that remains is the overshoot, so momentum is pure cost — 92 iterations against 36.

The property that decides this is the **condition number** $\kappa = \lambda_{\max}/\lambda_{\min}$ of the Hessian. Momentum’s convergence rate improves roughly from $1 - 2/\kappa$ to $1 - 2/\sqrt{\kappa}$, which is a large gain when κ is large and nothing at all when $\kappa = 1$. Our measurements agree: no help at $\kappa = 1$, $7.1\times$ at $\kappa = 200$, $12.5\times$ at $\kappa = 470$.

9. The two sources use different momentum conventions. PyTorch uses $v \leftarrow \beta v + g$, so the velocity is about $1/(1-\beta) = 10$ times a single gradient. Ng uses $v \leftarrow \beta v + (1-\beta)g$, a true running average, so the velocity is about the size of one gradient. Carrying $\eta = 0.001$ across unchanged therefore shrinks the effective step by a factor of $1-\beta = 0.1$, which is precisely the tenfold slowdown observed. The fix is $\eta = 0.01$ in the Ng convention. There is no bug in either formula; the bug is copying a hyperparameter between two definitions of the same symbol.

10. (a) By Equation (4) the largest usable learning rate is $2/L$, and L is the largest eigenvalue of the Hessian. Scaling one feature by 1000 scales its contribution to $X^\top X$ by 10^6 , so L rises by roughly that factor and $2/L$ falls correspondingly — measured, from 0.2485 to 4.42×10^{-4} , a factor of 562. (b) The convergence rate of gradient descent with an optimal η is $(\kappa-1)/(\kappa+1) \approx 1 - 2/\kappa$, so the number of iterations to a fixed accuracy grows proportionally to κ . Going from $\kappa = 470$ to $\kappa = 1.17 \times 10^8$ multiplies the iteration count by about 250,000.

A run of a few hundred iterations now takes tens of millions: the difference between a model that trains and one that does not. Note that the closed-form solution is unaffected by scaling — the harm is done entirely to the *optimiser*, which is exactly what makes this an optimisation decision rather than a cosmetic one.

13 References and further reading

All links were checked on 18 August 2026. Free resources are marked [free].

Textbooks

- Zhang, Lipton, Li and Smola, *Dive into Deep Learning*. The primary reference for this lecture. [free] §12.4 Stochastic Gradient Descent, §12.5 Minibatch Stochastic Gradient Descent, §12.6 Momentum. Every section is runnable in the browser. https://d2l.ai/chapter_optimization/sgd.html
https://d2l.ai/chapter_optimization/minibatch-sgd.html
https://d2l.ai/chapter_optimization/momentum.html
- Deisenroth, Faisal and Ong, *Mathematics for Machine Learning*, Cambridge 2020. [free] Chapter 7, and §7.1 in particular (“Optimization Using Gradient Descent”), which derives the step-size conditions and the momentum recursion used here. <https://mml-book.github.io/>
- Goodfellow, Bengio and Courville, *Deep Learning*, MIT Press 2016. [free] Chapter 8, “Optimization for Training Deep Models” — §8.1 on why batch methods fail at scale, §8.3 on SGD and momentum. <https://www.deeplearningbook.org/contents/optimization.html>
- Murphy, *Probabilistic Machine Learning: An Introduction*, MIT Press 2022. [free] Chapter 8, “Optimization”, for the convergence-rate statements quoted in Solution 8. <https://probml.github.io/pml-book/book1.html>
- Bishop, *Pattern Recognition and Machine Learning*, Springer 2006. §5.2.4 (gradient descent optimisation) and §5.4. A free PDF is hosted by Microsoft Research; their site blocks automated link checks, so no direct URL is printed here — search for the title plus “Microsoft Research”.

Online courses

- **NPTEL, Introduction to Machine Learning** — Prof. Balaraman Ravindran, IIT Madras. The closest match to this syllabus overall. [free] <https://nptel.ac.in/courses/106106139>
- **NPTEL, Deep Learning — Part 1** — Prof. Mitesh M. Khapra, IIT Madras. Week 4 covers exactly this lecture — gradient descent, momentum-based gradient descent, Nesterov accelerated gradient, stochastic and mini-batch gradient descent. [free] <https://archive.nptel.ac.in/noc/courses/noc19/SEM2/noc19-cs85/>
- **SWAYAM** — the national portal hosting NPTEL and other Indian university courses, with certification cycles. [free] <https://swayam.gov.in/>
- **Google Machine Learning Crash Course**, “Linear regression: Hyperparameters”. Covers learning rate, batch size and epochs with interactive widgets, and is the gentlest introduction on this list. [free] <https://developers.google.com/machine-learning/crash-course/linear-regression/hyperparameters>

Videos

- 3Blue1Brown, *Gradient descent, how neural networks learn / Deep Learning Chapter 2*. The best available visual account of what a gradient is and what descending it means. [free] <https://youtu.be/IHZwWFHwa-w>

- StatQuest, *Gradient Descent, Step-by-Step*. Works a complete two-parameter example arithmetically, at a deliberate pace. [free] <https://youtu.be/sDv4f4s2SB8>
- StatQuest, *Stochastic Gradient Descent, Clearly Explained!!!*. The follow-up, on why sampling the sum works. [free] <https://youtu.be/vMh0zPT0tLI>
- DeepLearningAI (Andrew Ng), *Gradient Descent With Momentum (C2W2L06)*. Nine minutes, and the source of the ravine picture everyone uses. Watch *Exponentially Weighted Averages (C2W2L03)*, <https://youtu.be/1Aq96T8FkTw>, first if the $1/(1 - \beta)$ factor is not yet obvious. Note that Ng uses the $(1 - \beta)$ convention discussed in Section 8. [free] https://youtu.be/k8fTYJPd3_I
- StatQuest video index — short explanations for most topics in this syllabus. [free] https://statquest.org/video_index.html

Documentation and interactive explainers

- Goh, “Why Momentum Really Works”, *Distill*, 2017. An interactive article with sliders for η and β over a ravine; twenty minutes here is worth a chapter of reading. [free] <https://distill.pub/2017/momentum/>
- scikit-learn, 1.5. *Stochastic Gradient Descent*. The user guide, including the practical-tips section on scaling and on choosing `eta0`. [free] <https://scikit-learn.org/stable/modules/sgd.html>
- scikit-learn, `SGDRegressor` API reference — the arguments named in Section 9. [free] https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.SGDRegressor.html
- PyTorch, `torch.optim.SGD`. Read the “Note” block: it states the exact momentum convention used, which is Equation (8). [free] <https://pytorch.org/docs/stable/generated/torch.optim.SGD.html>

Articles

- Ruder, “An overview of gradient descent optimization algorithms”, arXiv:1609.04747, 2016. Sections 1–4 cover exactly this lecture; the rest is L4. [free] <https://arxiv.org/abs/1609.04747>
- Bottou, “Large-Scale Machine Learning with Stochastic Gradient Descent”, COMPSTAT 2010. Six pages, and the clearest statement of why SGD’s noise is an acceptable price at scale. [free] <https://leon.bottou.org/publications/pdf/compstat-2010.pdf>
- Polyak, “Some methods of speeding up the convergence of iteration methods”, *USSR Computational Mathematics and Mathematical Physics* 4(5), 1964. The original heavy-ball paper, and the source of Equation (8).

Sources: Zhang, Lipton, Li and Smola, *Dive into Deep Learning*, Ch. 12 (Optimization Algorithms); Deisenroth, Faisal and Ong, *Mathematics for Machine Learning*, Ch. 7.; Kingma and Ba, “Adam: A Method for Stochastic Optimization,” ICLR 2015.; Ruder, “An overview of gradient descent optimization algorithms,” arXiv:1609.04747, 2016.